

Google I/O predictions

According to an analyst, expect to hear a lot about wearables and design at this month's Google I/O event <http://dgit.in/TGCb1N>



Samsung's new watch

Reports are pouring in about the South Korean giant's new smartwatch, and it doesn't need a smartphone <http://dgit.in/1osfw3i>

highest density of 441ppi. The Super AMOLED display on the Galaxy S5, in our opinion, is one of the best displays that we've seen in a long time. The colours and brightness levels on the display are excellent and viewing angles are also great. The display on the Xperia Z2 has improved a lot from the Z1 as Sony has finally put in an IPS panel making the viewing angles much better. With technologies like X-Reality, Triluminos and Live Colour LED, the display is actually quite crisp and offers true colours. The display on the HTC One (M8) is quite bright and offers excellent brightness levels. Out of the three devices, the One (M8) offers the best touch response and the display feels silky smooth.



Does super-slim = super-awesome?

The two major hardware updates that were seen this year were the addition of a heart-rate monitor and a fingerprint scanner on the Galaxy S5. Now we know that HTC and Apple have already incorporated a fingerprint scanner in their devices, but this is the first ever PayPal approved scanner. This means that you can perform PayPal transactions by just swiping your finger over the scanner. The heart-rate monitor is placed at the back right under the camera where you need to

keep your finger to determine your heart rate. Both the features work quite well and Samsung deserves a pat on the back for the new hardware additions.

Sony hasn't made any major hardware changes and stands apart with its simpler design language. The only major and welcoming hardware change we noticed was the new loudspeaker. The Z1 and the Z1 compact had a loudspeaker at the bottom, which not only lacked volume. The Z2 now has dual-stereo speakers on either sides

of the display which seems like the perfect placement.

The newest addition from HTC this year has been the new duo-lens camera with a dual-tone LED flash. Having a dual-lens system allows you take pictures and instantly refocus on the area you want. Also the new dual-tone LED flash fires a warm or a cool beam of light depending on the colour temperature of the subject(s).

User Interface

All the three smartphones run on the latest Android 4.4.2 KitKat update and apart from the usual Android goodness, each smartphone offers a personalised experience thanks to the customized UI that's layered atop the core OS.

Samsung's TouchWiz UI has evolved over the years and even today it offers a vibrant and colourful experience. This is primarily to make full use of the Super AMOLED display. The new TouchWiz UI brings in a new lockscreen, notification bar with quick toggles and of course new apps. Samsung has also integrated a Flipboard based app called My Magazine which can be easily accessed with a swipe right from the first home-screen. There is also a health oriented app called S-Health which keeps a track of your workout and food habits. The system menu has also been overhauled with new round icons. The new TouchWiz does look pretty, but we wish it was a little smoother and bug free.

Sony's UI is simple and minimalist with good looking icons and some useful apps. The one thing that we love about Sony is that they don't make a mess of their UI and offers something which doesn't take a lot of time to get used to. Sony also doesn't have a lot of bloat-ware which

HOW WE TESTED

Testing a smartphone isn't as simple as it seems. The sheer diversity makes direct comparisons very tough, and hinge on a lot of factors.

Performance

First off, we run a series of benchmarks to get a set of numbers allowing us to directly compare the performance of all the devices - processor, graphics, storage and even the web browser. Each platform largely has its own set of benchmarks, with the Browsermark 2.0 and the Octane tests being true cross platform ones. We have given equal weightage to all benchmark scores, so as to not put the other platforms at a disadvantage. These tests give us an idea of how the hardware performs, in isolation as well as comparatively.

Shifting to real-world use, call quality is of paramount importance in a smartphone, and we tested all phones in three different signal zones - full signal, 50% signal and very low signal, usually 1-2 bars. Points are reduced for every call drop, voice echo and audio break situations and also when audio is completely lost. To keep the testing on an even platform, the same SIM card from the same operator is used on all phones, so that any network variation does not have an impact on the results. Usability is critical, which is why we look at the user interface on the device, the quality of the on-screen keypad, variety of apps preloaded, etc. The build quality, materials used, placement of ports and keys, are other aspects that are tested and rated.

Features and Design

Multimedia, with the current generation of smartphones, gains a lot of importance. We have tests for display quality, viewing angles, performance of the speaker etc. We also test and rate displays on aspects like usability in sunlight.

The battery life test was done to see how long the phone lasts on a single charge. The usage included video playback, web browsing and IM chats, a dollop of Angry Birds and some voice calls, all with 3G and Wi-Fi connected throughout. The hardware and specifications do have considerable amount of weightage to them, and we have allocated points for everything like processor type, display type, etc., to ensure every little nuance is noted and rated accordingly